Justin Xu

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EDUCATION

Georgia Institute of Technology Atlanta, GA
Bachelor of Science in Computer Science Expected Graduation Date: May 2027
Concentrations: Intelligence and Modeling and Simulation

COURSEWORK

CS 1332 – Data Structures and Algorithms | CS 3600 – Intro to AI | CS 4400 – Intro to Database Systems | CS 4641– Machine Learning | CS 4251 – Computer Networking II | CS 4650 – Natural Language Processing | CS 3630 – Perception and Robotics | MATH 1554 – Linear Algebra | MATH 2551 – Multivar. Calculus | MATH 3670 – Statistics | MATH 3012 – Combinatorics

SKILLS

- **Programing Languages:** Python, Java, C++, C, SQL, Bash, R, HTML, CSS
- **Operating Systems/Platforms:** Windows, Linux, Unix, Azure, Kubernetes, Openshift, Docker, Raspberry Pi
- **Frameworks:** PyTorch, LangChain, RAG, Flask
- **Software/Packages/Tools:** NumPy, SciPy, Matplotlib, Pandas, Keras, Scikit-Learn, ChromaDB, MySQL, FAISS, Transformers, Jupyter Notebook, Git/Github, LAMMPS, Android Studio, Junit, Firebase, Ansible, VS Code, Cursor, Claude, Power BI

EXPERIENCE

Georgia Tech Vertically Integrated Projects Atlanta, GA
Undergraduate Researcher for the Surrogate Modeling for Urban Regeneration (SMUR) team January 2025 - May 2026

- Developed agent-based simulation software in Python that uses game theory and no-regret dynamics to simulate the demographic changes of the Atlanta Beltline.
- Used python modules such as threading, OS, requests, pandas, pickle, and matplotlib to fetch, store, cache, and visualize simulation data.

Georgia Tech EIC (Efficient and Intelligent Computing) Lab Atlanta, GA
Undergraduate Intern October 2025 – December 2025

- Developed scripts enabling improved model conversion and quantization pipeline for edge deployment of AI models.
- Analyzed caching mechanisms in Diffusion LLM models for potential improvements in performance and efficiency.

Georgia Tech OSPO (Open Source Program Office) Atlanta, GA
Summer Intern May 2025 – July 2025

- Worked with IBM engineers in troubleshooting and identifying key issues that prevented Ansible playbooks from running on z/OS.
- Contributed to the zopen open-source community solutions that allowed Ansible playbooks to run directly on z/OS, enabling automated configuration management and deployment right from automation pipelines.

Georgia Tech Bhamla Lab Atlanta, GA
Undergraduate Researcher June 2024 – May 2025

- Worked on a research project on active polymers using LAMMPS (Large-scale Atomic/Molecular Massively Parallel Simulator) Brownian physics simulation software.
- Automated the process of collecting, organizing, and visualizing simulation data using Python data analysis packages such as NumPy, SciPy, and Matplotlib.
- Research published in the journal Physical Review Letters (PRL): https://doi.org/10.1103/ntm6-dz86

Novelis Global Research and Development Center Kennesaw, GA
Research and Development Intern June 2023 – July 2023

- Worked with a team to develop a robotic autonomous system for transporting and administering a three-point bend test on various aluminum alloy samples.
- Used python modules such as threading, OS, sys, and RPi.GPIO libraries to control step motors simultaneously using pulse width modulation (PWM) signals and to program general purpose input/output (GPIO) pins to receive and send signals between various hardware, such as limit switches, test machinery, and a UR16e robotic arm.